

Badre Mhiouah

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Quantitative Finance & Data Science

SUMMARY

Quantitative developer with 1.5 years experience in derivatives trading (XVA, gap risk) and AI-driven risk analytics at a bulge-bracket bank. Proven track record delivering trading strategies, risk models and AI automation tools, seeking to contribute to Balyasny Asset Management's pod-centric quant research environment.

SKILLS & INTERESTS

Agentic AI:	LLM Applications, Agentic AI, LangGraph, Tool Calling, MCP, Multi-Agent Systems, Context Engineering, LangChain, LLM APIs
RAG & Evaluation:	RAG, Embeddings, Vector Search, Hybrid Search, Reranking, LLM Evaluation, Ragas, DeepEval, Observability
AI Engineering:	Python, SQL, FastAPI, REST APIs, Docker, Git, CI/CD, AWS/GCP, Monitoring.
ML & Fine-Tuning:	PyTorch, Hugging Face, Transformers, LoRA/QLoRA, PEFT, Deep Learning, Data mining, Inference, Prediction
Data:	Pandas, NumPy, Scikit-learn, Experiment Design, Statistical Testing
Languages:	French, Arabic, English (fluent); Spanish (intermediate); Mandarin (elementary)

EXPERIENCE

Societe Generale Corporate and Investment Banking : Data Scientist	Jun.2025 – Jul.2026
<i>General Inspection</i>	<i>Hong Kong</i>
- Partnered with non-technical stakeholders to identify time-consuming, repetitive audit activities and assess opportunities for automation, translating business needs into data and AI use cases. - Designed and prototyped LLM-based solutions using prompt engineering and LLM APIs to automate and accelerate audit workflows, including the enrichment, structuring, and analysis of operational loss data. - Developed a Power BI platform combining automated insights and multi-level risk reporting, enabling auditors to reduce manual analysis and gain faster access to actionable risk information. - Analyzed large-scale trading datasets from Delta One activities to identify anomalous patterns, risk concentrations, and control weaknesses across equity and index derivative products, supporting auditors in risk-focused investigations. - Performed forensic analyses of trading and market data to identify potential indicators of market manipulation. - Delivered Python and Power BI training to auditors, enabling self-service analytics and facilitating the adoption of data-driven approaches within the IGAD teams.	
Barclays : Trader intern	Aug.2023 – Jan. 2024
<i>Cross-asset derivatives trading desk</i>	<i>Paris & London</i>
- Developed a trading strategy aimed at hedging own credit risk, resulting in a 7% boost in profitability and a 5% decrease in P&L volatility. (Bloomberg, Python, Machine learning) - Improved risk management instruments to pinpoint counterparties with significant impacts on each XVA by country, currency and greeks. (Excel, VBA, API) - Modeled and calculated the effects of gap risk on both credit and own credit risks for the portfolio of the bank. - Developed reporting tools to regularly analyze gap risk, facilitating the identification of its key drivers. - Collected and cleaned interest rate data for major currencies, automated data processing, and developed optimal neural network models for yield curve prediction with backtesting and hyperparameter tuning.	

PERSONAL PROJECTS (AVAILABLE ON MY GITHUB)

Financial News Sentiment & Market Analysis Pipeline	Aug. 2026 – Present
<i>Python, Hugging Face, FinBERT, FastAPI, PostgreSQL, Docker</i>	
Goal: Building an end-to-end AI pipeline to extract sentiment from financial news, transform it into quantitative signals, and evaluate its relationship with subsequent stock market movements. AI & Data Pipeline: Collecting financial news and historical market data, processing and storing them in a database, and using FinBERT for domain-specific sentiment analysis and continuous sentiment scoring. Quantitative Analysis: Aggregating sentiment into trading signals and implementing simple trading strategies and backtests to evaluate predictive power, returns, and risk against baseline approaches.	

- **AI Engineering:** Developing a modular Python application and progressively exposing its inference and analysis capabilities through **FastAPI**, with **PostgreSQL** and **Docker** planned for scalable data management and reproducible deployment.

Jane Street Puzzle – August 2026

Aug. 2026

Mathematics, Python, LLMs, Cursor

- **AI-Assisted Reasoning:** Using multiple frontier LLMs and **Cursor** to explore mathematical approaches, generate code and challenge proposed solutions.
- **Computational Validation:** Developing Python simulations and visualizations to test mathematical intuitions and validate analytical results.

EDUCATION

CentraleSupélec

Sep. 2020 – Apr. 2023

Engineering Degree, Major in Mathematical Modelling & Data Science, Minor in Financial Mathematics *Paris, France*

- Relevant coursework: Mathematics, Statistics, Machine Learning, Optimization, Stochastic Models, Financial Modeling, Risk Management, Derivatives, Fixed Income, Economics, Negotiation course, Data structures, Algorithms & Complexity.

Université Côte d’Azur

Sep 2017 – Apr. 2020

Bachelor’s degree in Advanced Mathematics

Nice, France

- Main courses: Topology, Complex analysis, Numerical analysis, Differential calculus, Algebra & Geometry.

EXTRACURRICULAR ACTIVITIES

Gap Year – Independent Travel (Southeast Asia)

Jan. 2024 – Jan. 2025

Independent Travel

Southeast Asia

- Undertook a year of independent travel across Southeast Asia, gaining cross-cultural experience and adaptability while navigating diverse environments and logistical challenges.
- Developed strong self-management, problem-solving, and communication skills through extended travel in countries including Thailand, Vietnam, Cambodia, Indonesia.
- Exposure to different cultural, economic, and social contexts strengthened intercultural awareness and resilience.